

Amber Li

amberli2@andrew.cmu.edu • (434) 851-7666 • [amburger66.github.io](https://github.com/amburger66)

EDUCATION

Carnegie Mellon University

Doctor of Philosophy in Robotics (GPA: 5.0/5.0)

Pittsburgh, PA

Expected: May 2029

Massachusetts Institute of Technology

Master of Engineering in Computer Science: concentration in Artificial Intelligence (GPA: 5.0/5.0)

Cambridge, MA

February 2023

Bachelor of Science in Computer Science, minor in Mathematics (GPA: 4.9/5.0)

May 2022

PUBLICATIONS

Kallol Saha*, **Amber Li***, Ángela Rodríguez-Izquierdo*, Lifan Yu, Ben Eisner, Maxim Likhachev, David Held. Planning from Point Clouds over Continuous Actions for Multi-object Rearrangement. Conference on Robot Learning (CoRL) 2025. **Oral presentation (Top 5.7%)**.

Amber Li, Tom Silver. Embodied Active Learning of Relational State Abstractions for Bilevel Planning. Conference on Lifelong Learning Agents (CoLLAs) 2023. **Oral presentation (Top 12)**.

SKILLS

Programming: Python, Java, C

Frameworks: PyTorch, NumPy, Pytorch3d, PyBullet, ManiSkill, Pandas, Git, LaTeX

Languages: English (native), Mandarin Chinese (conversant)

RESEARCH EXPERIENCE

Robots Perceiving and Doing Lab, Search-Based Planning Lab

Pittsburgh, PA

Graduate Student Researcher (Advisors: David Held & Maxim Likhachev)

September 2024 – Present

- Research focus: integrating deep learning and classical planning for long-horizon robotic manipulation tasks

Learning and Intelligent Systems Group

Cambridge, MA

Graduate Student Researcher (Advisor: Leslie Kaelbling)

September 2021 – January 2023

- Thesis: Embodied Active Learning of Relational State Abstractions for Bilevel Planning
- Contributed to a framework for bilevel planning using learned neuro-symbolic relational abstractions by proposing an embodied active learning paradigm for learning state abstractions
- Implemented an online algorithm that outperforms all baselines while exhibiting sample efficiency on planning tasks in three robotic environments

Computer-Aided Programming Group

Cambridge, MA

Undergraduate Student Researcher (Advisor: Armando Solar-Lezama)

December 2020 – May 2021

- Explored modeling types of political governance systems with the goal of inferring an optimal system given desired societal outcomes; leveraged techniques from reinforcement learning and program synthesis

Marin Soljačić Group

Cambridge, MA

Undergraduate Student Researcher (Advisor: Marin Soljačić)

September 2020 – May 2021

- Constructed a graph neural network model that can be trained end-to-end and used to extract interpretable equations, such as underlying mathematical laws, from data
- Demonstrated the advantages of this architecture for symbolic regression over simpler approaches

PROFESSIONAL EXPERIENCE

Two Sigma Investments

New York, NY

Software Engineer

February 2023 – June 2024

- Built and migrated data pipelines, developed infrastructure and data profiling tools, and contributed operational support as part of the Modeling Data Engineering team

TEACHING EXPERIENCE

Massachusetts Institute of Technology

Cambridge, MA

Introduction to Algorithms | *Graduate Teaching Assistant*

Spring 2023, Fall 2023

- Taught recitation sections (2x/week), managed all graders for grading biweekly problem sets, wrote recitation notes and problem sets, staffed office hours, graded exams

HAUSCR Summit for Young Leaders in China

Virtual

How to Give Presentations | *Core Lecturer*

August 2020

LEADERSHIP AND ACTIVITIES

MIT Women in EECS | *Mentorship Committee Member, Co-President* September 2019 – May 2022

- Elected co-president of student-run organization that seeks to support and empower women in EECS by building community
- Led bi-weekly exec meetings, collaborated with EECS admin, drove the creation of the club's first corporate partnership program
- Organized mentor-mentee matches and planned monthly events for the 100+ member mentorship program

Society of Women Engineers (SWE) – MIT Chapter | *Technology Chair* February 2019 – December 2020

- Organized and taught technology workshops to SWE members and the MIT community
- Mentored middle school girls in web development using HTML/CSS and JavaScript

ACTIVITIES & AWARDS

CMU Robotic Vision Fellowship | *Awardee* October 2025

CMU Manipulation Discussion Group | *Organizer* September 2025 – Present

CMU All-University Orchestra | *First Violinist* September 2024 – Present

HKN (Eta Kappa Nu, EECS Honor Society) – MIT Chapter | *Member, Tutor* May 2020 – May 2022

MIT Chamber Music Society | *Violinist, Chamber Music Performance Awardee* September 2019 – December 2022

SERVICE

Conference reviewing: CoRL 2025